

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	257375	53.225700
1	[Git, Change, Create]	74452	15.396833
2	[Git, Change, Delete]	44503	9.203315
3	[Git, Change, Rename]	23284	4.815181
4	[File, Git]	15911	3.290429
5	[Git, Commit]	15582	3.222391
6	[Java, ByteCode, Parameter]	9227	1.908163
7	[Java, ByteCode, Member, Method]	8882	1.836817
8	[Java, ByteCode, Bound]	6300	1.302853
9	[Java, Value, ByteCode, Annotation]	5334	1.103083
10	[Java, ByteCode, Bound, ParameterizedType]	4849	1.002784
11	[Xml, Element]	2384	0.493016
12	[Java, ByteCode, Member, Field]	2187	0.452276
13	[Xml, Text]	1596	0.330056
14	[Java, ByteCode, Member, Constructor, Method]	1583	0.327368
15	[Java, ByteCode, Bound, WildcardType]	968	0.200184
16	[Java, ByteCode, Member, Method, Lambda]	851	0.175989
17	[Java, ByteCode, Bound, TypeVariable]	830	0.171646
18	[Type, File, Java, ByteCode, JavaType]	689	0.142487
19	[Type, File, Java, Class, ByteCode]	647	0.133801
20	[Json, Key]	533	0.110226
21	[Value, Json, Scalar]	523	0.108158
22	[Git, Change, Copy]	470	0.097197
23	[Type, File, Java, ByteCode, ResolvedDuplicate...	459	0.094922
24	[Java, ByteCode, GenericDeclaration, Member, M...	455	0.094095
25	[Java, Value, ByteCode, Primitive]	322	0.066590
26	[Author, Git, Person]	302	0.062454
27	[Type, File, Java, ByteCode, ExternalType]	301	0.062247
28	[Committer, Git, Person]	246	0.050873
29	[Maven, Dependency]	192	0.039706

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

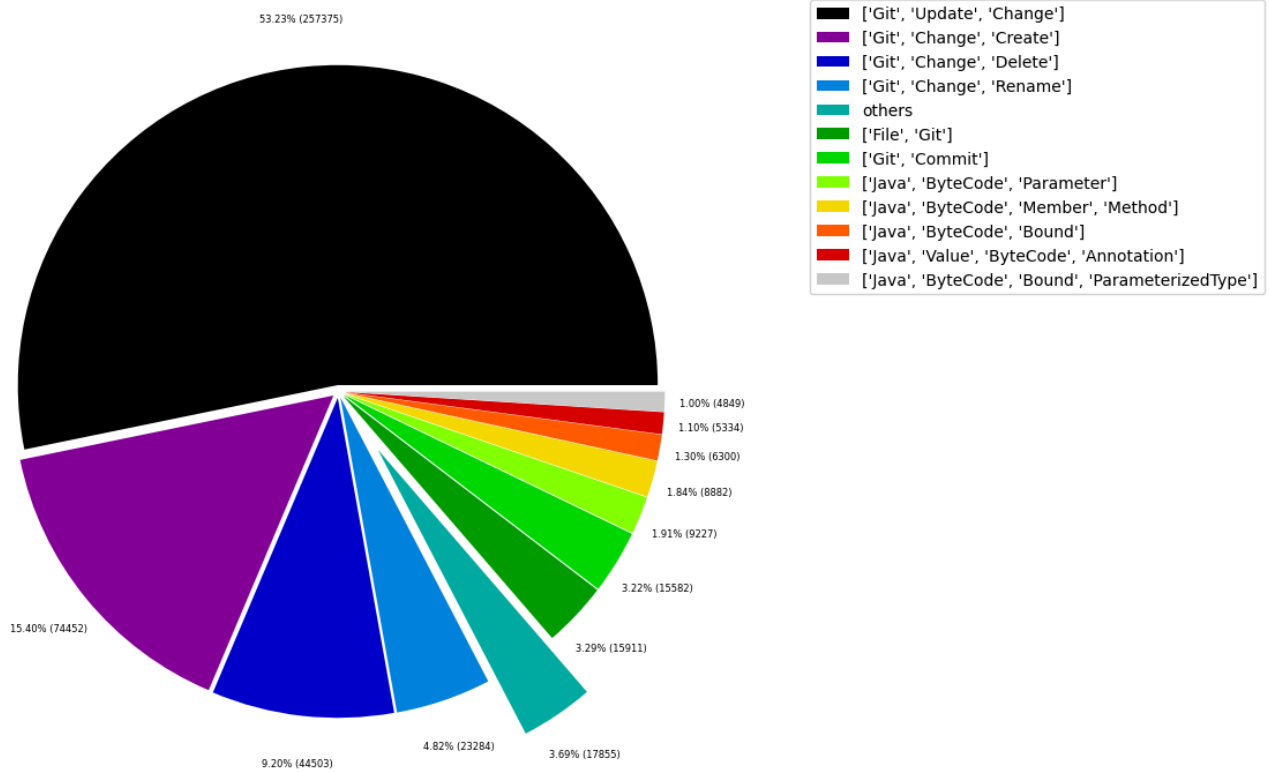


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000207
1	[Git, Branch]	1	0.000207
2	[Repository, File, Git]	1	0.000207
3	[File, Json]	2	0.000414
4	[File]	3	0.000620
5	[Maven, Exclusion]	4	0.000827
6	[Type, File, Java, ByteCode, GenericDeclaratio...	4	0.000827
7	[Value, Array, Json]	8	0.001654
8	[Type, File, Java, ByteCode, Throwable, Extern...	8	0.001654
9	[Java, ByteCode, GenericDeclaration, Member, C...	9	0.001861
10	[Maven, Scm]	11	0.002275
11	[File, Maven, Xml, Pom, Document]	11	0.002275
12	[Type, File, Java, ByteCode, Throwable, Resolv...	11	0.002275
13	[Type, File, Java, ByteCode, Void]	11	0.002275
14	[Java, ManifestSection]	11	0.002275
15	[File, Java, Manifest]	11	0.002275
16	[File, Artifact, Jar, Archive, Zip, Java]	11	0.002275
17	[File, Java, ServiceLoader]	12	0.002482
18	[Maven, PluginExecution]	13	0.002688
19	[File, Java, Properties]	13	0.002688
20	[Maven, ExecutionGoal]	13	0.002688
21	[Type, File, Java, ByteCode, Enum]	17	0.003516
22	[jQAssistant, Rule, Concept]	19	0.003929
23	[Maven, Configuration]	20	0.004136
24	[Maven, Plugin]	20	0.004136
25	[Xml, Attribute]	22	0.004550
26	[Type, File, Java, ByteCode, PrimitiveType]	39	0.008065
27	[Type, File, Java, ByteCode, Annotation]	42	0.008686
28	[Type, File, Java, Class, ByteCode, Throwable]	42	0.008686
29	[Xml, Namespace]	44	0.009099

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

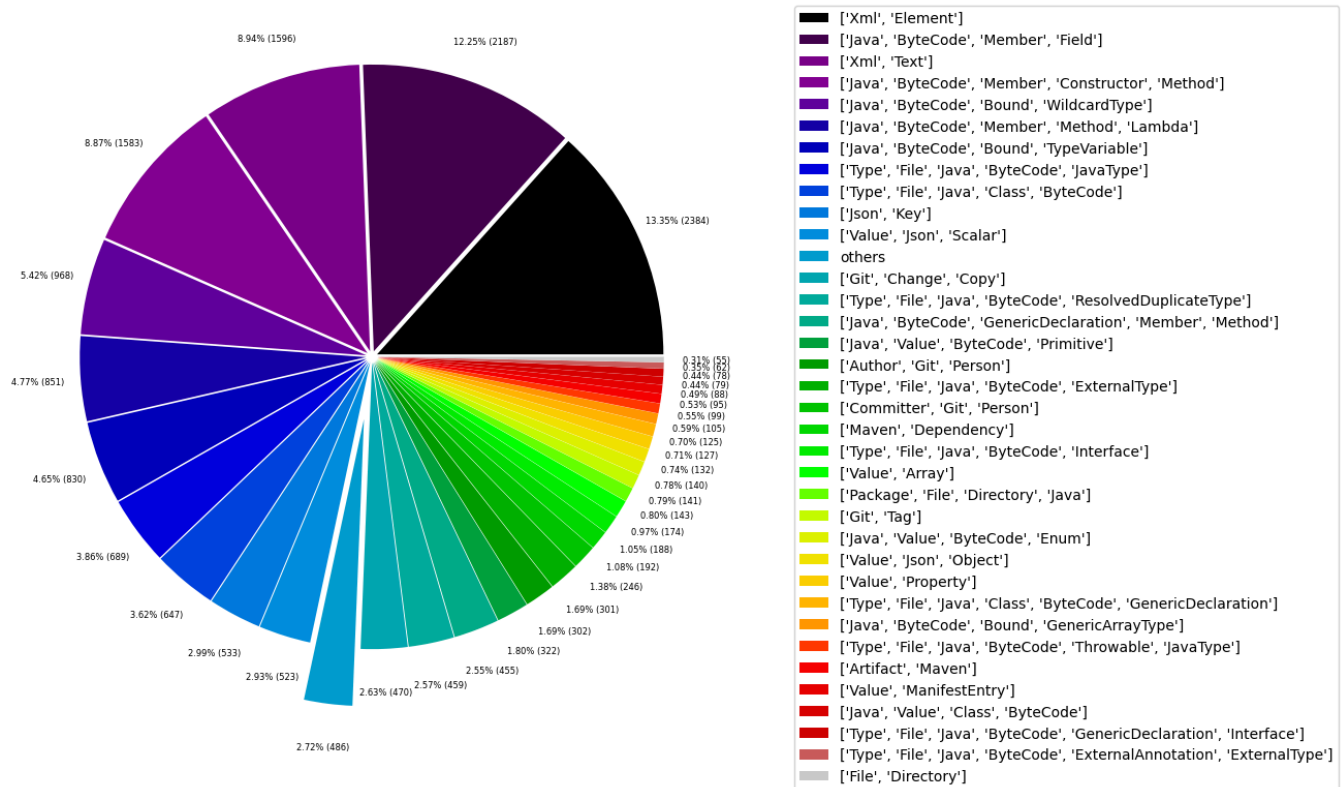


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

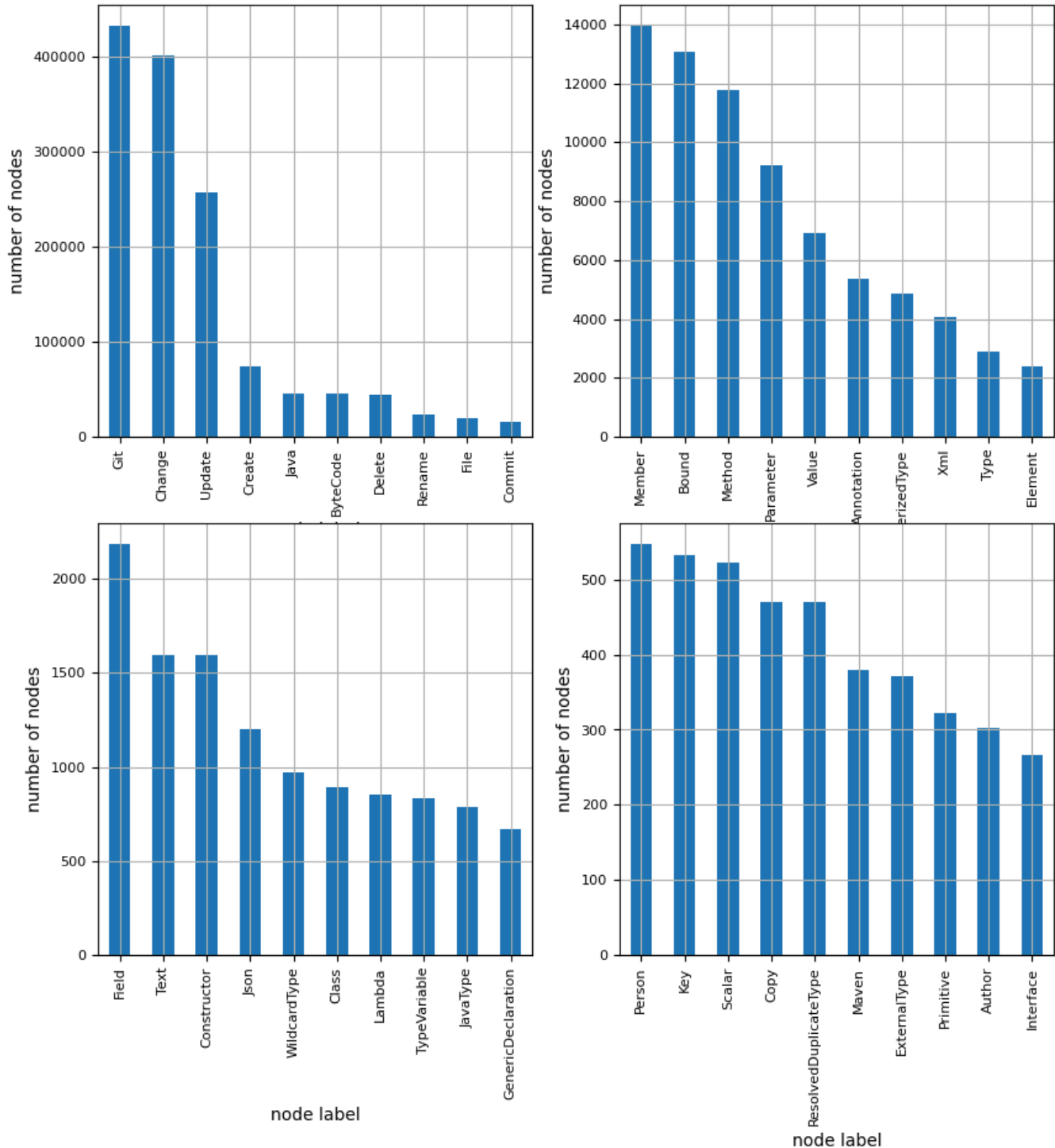
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	432268	89.393946
1	Change	400084	82.738226
2	Update	257375	53.225700
3	Create	74452	15.396833
4	Java	45196	9.346629
5	ByteCode	44995	9.305062
6	Delete	44503	9.203315
7	Rename	23284	4.815181
8	File	19047	3.938960
9	Commit	15582	3.222391
10	Member	13967	2.888405
11	Bound	13052	2.699181
12	Method	11780	2.436129
13	Parameter	9227	1.908163
14	Value	6927	1.432518
15	Annotation	5376	1.111768
16	ParameterizedType	4849	1.002784
17	Xml	4057	0.838996
18	Type	2874	0.594349
19	Element	2384	0.493016
20	Field	2187	0.452276
21	Text	1596	0.330056
22	Constructor	1592	0.329229
23	Json	1198	0.247749
24	WildcardType	968	0.200184
25	Class	893	0.184674
26	Lambda	851	0.175989
27	TypeVariable	830	0.171646
28	JavaType	788	0.162960
29	GenericDeclaration	671	0.138764
30	Person	548	0.113328
31	Key	533	0.110226
32	Scalar	523	0.108158
33	Copy	470	0.097197
34	ResolvedDuplicateType	470	0.097197
35	Maven	379	0.078378
36	ExternalType	371	0.076724
37	Primitive	322	0.066590
38	Author	302	0.062454
39	Interface	266	0.055009

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1501861

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	400084	26.639216
1	MODIFIES	400084	26.639216
2	UPDATES	257375	17.137072
3	CREATES	98206	6.538954
4	DELETES	67787	4.513534
5	COMMITTED	31164	2.075026
6	RENAMES	23284	1.550343
7	INVOKES	22207	1.478632
8	HAS_PARENT	18926	1.260170
9	OF_TYPE	18495	1.231472
10	DEPENDS_ON	17561	1.169283
11	HAS_FILE	15911	1.059419
12	HAS_COMMIT	15582	1.037513
13	DECLARES	14306	0.952552
14	OF_RAW_TYPE	13122	0.873716
15	HAS	9843	0.655387
16	HAS_NEW_NAME	9378	0.624425
17	RETURNS	8100	0.539331
18	READS	5929	0.394777
19	HAS_ACTUAL_TYPE_ARGUMENT	5753	0.383058
20	ANNOTATED_BY	5333	0.355093
21	OF_GENERIC_TYPE	4392	0.292437
22	CHANGED_TOGETHER_WITH	3686	0.245429
23	SIMILAR	3370	0.224388
24	CONTAINS	3183	0.211937
25	RETURNS_GENERIC	2586	0.172186
26	RESOLVES_TO	2573	0.171321
27	HAS_FIRST_CHILD	2384	0.158736
28	HAS_LAST_CHILD	2384	0.158736
29	HAS_ELEMENT	2362	0.157272

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

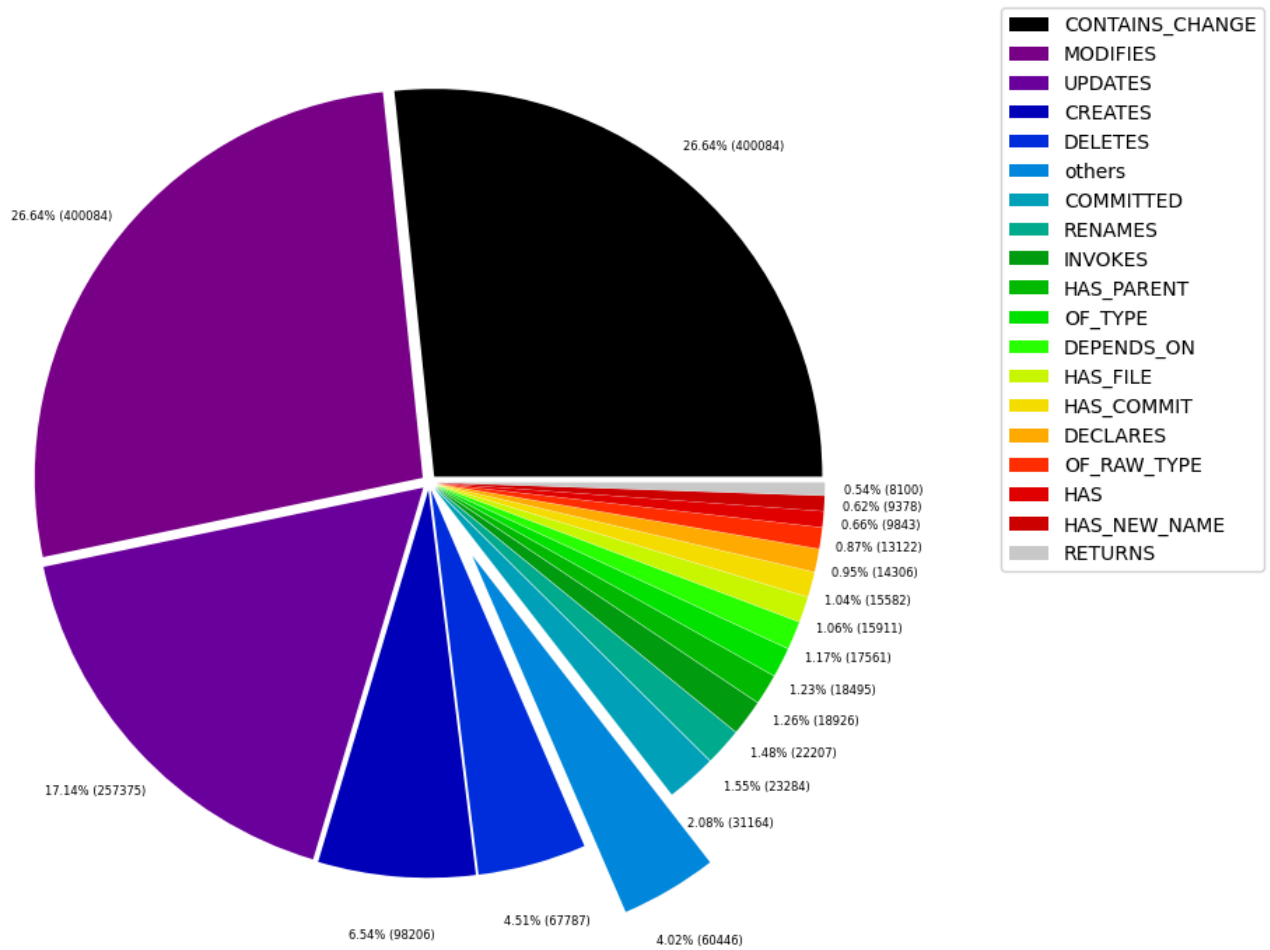


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

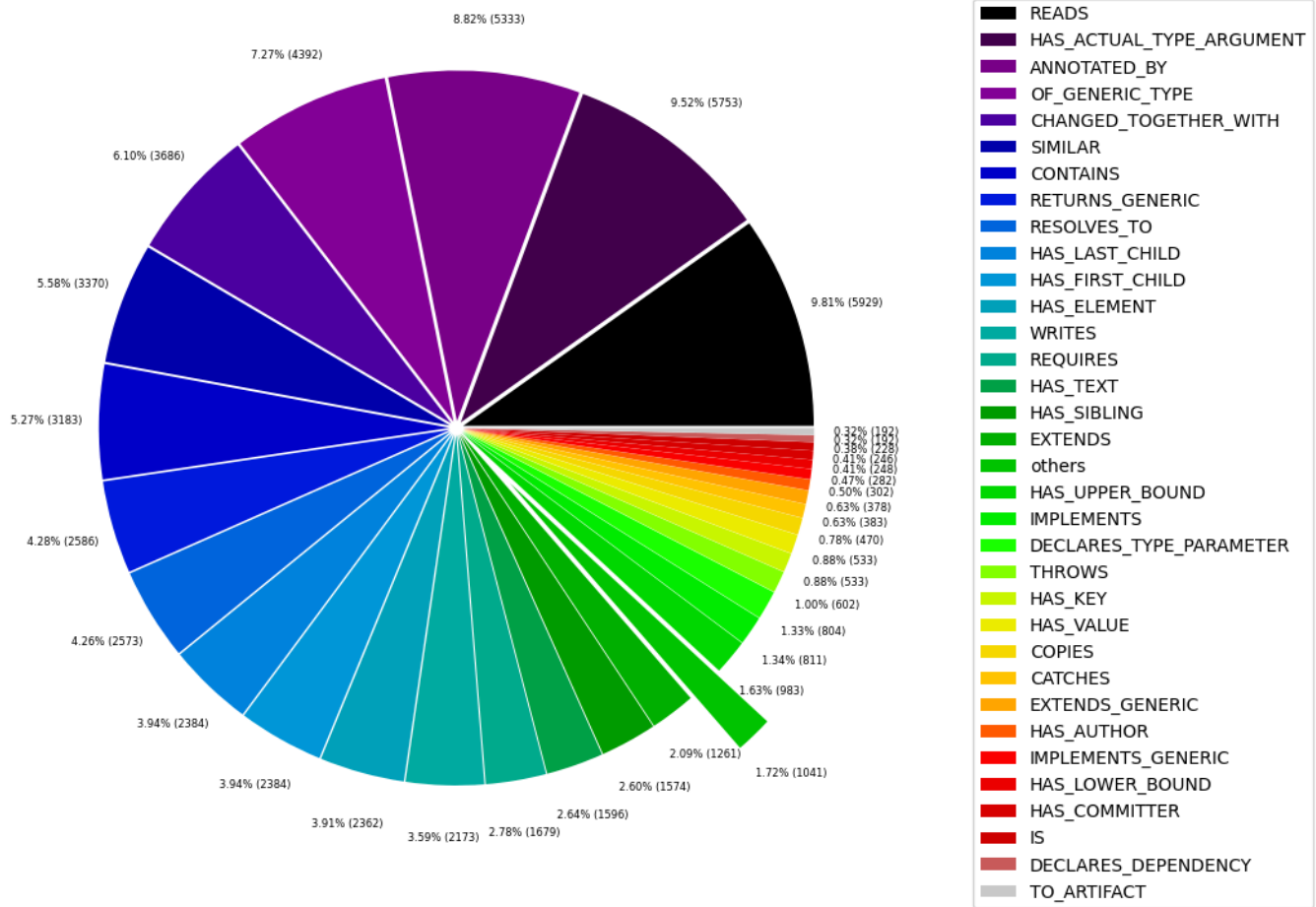
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000067
1	HAS_BRANCH	1	0.000067
2	HAS_HEAD	2	0.000133
3	THROWS_GENERIC	3	0.000200
4	EXCLUDES	4	0.000266
5	HAS_SCM	11	0.000732
6	DESCRIBES	11	0.000732
7	HAS_ROOT_ELEMENT	12	0.000799
8	HAS_GOAL	13	0.000866
9	HAS_EXECUTION	13	0.000866
10	INCLUDES_CONCEPT	19	0.001265
11	IS_ARTIFACT	20	0.001332
12	HAS_CONFIGURATION	20	0.001332
13	USES_PLUGIN	20	0.001332
14	OF_NAMESPACE	22	0.001465
15	HAS_ATTRIBUTE	22	0.001465
16	REQUIRES_TYPE_PARAMETER	26	0.001731
17	REQUIRES_CONCEPT	28	0.001864
18	DECLARES_NAMESPACE	44	0.002930
19	HAS_DEFAULT	53	0.003529
20	HAS_COMPONENT_TYPE	105	0.006991
21	CONTAINS_VALUE	128	0.008523
22	HAS_TAG	141	0.009388
23	ON_COMMIT	141	0.009388
24	COPY_OF	181	0.012052
25	TO_ARTIFACT	192	0.012784
26	DECLARES_DEPENDENCY	192	0.012784
27	IS	228	0.015181
28	HAS_COMMITTER	246	0.016380
29	HAS_LOWER_BOUND	248	0.016513

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberO
0	[Git, Update, Change]	UPDATES	[File, Git]	257375		257375
1	[Git, Update, Change]	MODIFIES	[File, Git]	257375		257375
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	257375		15582
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	74452		15582
4	[Git, Change, Create]	MODIFIES	[File, Git]	74452		74452
5	[Git, Change, Create]	CREATES	[File, Git]	74452		74452
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	44503		15582
7	[Git, Change, Delete]	MODIFIES	[File, Git]	44503		44503
8	[Git, Change, Delete]	DELETES	[File, Git]	44503		44503
9	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	23284		15582
10	[Git, Change, Rename]	RENAMES	[File, Git]	23284		23284
11	[Git, Change, Rename]	CREATES	[File, Git]	23284		23284
12	[Git, Change, Rename]	DELETES	[File, Git]	23284		23284
13	[Git, Change, Rename]	MODIFIES	[File, Git]	23284		23284
14	[Git, Commit]	HAS_PARENT	[Git, Commit]	18915		15582
15	[Repository, File, Git]	HAS_FILE	[File, Git]	15911		1
16	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	15582		1
17	[Committer, Git, Person]	COMMITTED	[Git, Commit]	15582		246
18	[Author, Git, Person]	COMMITTED	[Git, Commit]	15582		302
19	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	13259		8857
20	[File, Git]	HAS_NEW_NAME	[File, Git]	9378		15911
21	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	5332		8857
22	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	5325		8857
23	[Java, Value, ByteCode, Annotation]	OF_TYPE	[Type, File, Java, ByteCode, ExternalAnnotatio...	4990		5334
24	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, javaType]	4105		9227
25	[Java, ByteCode, Parameter]	ANNOTATED_BY	[Java, Value, ByteCode, Annotation]	3797		9227
26	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	3068		15911
27	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	2799		4849
28	[Xml, Element]	HAS_ELEMENT	[Xml, Element]	2362		2384
29	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	2256		6300

Graph Density

```
total_number_of_nodes (vertices): 483554
total_number_of_relationships (edges): 1501861
-> total directed graph density: 6.42304075361977e-06
-> total directed graph density in percent: 0.000642304075361977
```